

A Comparative Performance Study on Recurrent Time to Event Data

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DECLARATION

This is to certify that the STAT1092-Project titled “**A Comparative Performance Study on Recurrent Time to Event Data**” has been done by the students **SHAGNICK ROY**, Reg. No.: **21214110023** of 2021-22 and **PRAPTI PODDAR**, Reg. No.: **21214210004** of 2021-22, M.Sc. 2nd Year, Semester 4 of the Department of Statistics, Presidency University, Kolkata during the academic year **2025-26**. They have worked independently under my guidance and completed it to my satisfaction.

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1 Introduction

In survival studies, we often come across events that are not terminal, i.e., they repeat themselves after certain time intervals. For example, patients diagnosed with chronic obstructive pulmonary disease (COPD) often report the recurrence of symptoms after a period of time. After a treatment has been registered, the patient remains in a state of wellness, followed by reemergence of the disease and a state of illness. This cycle continues to repeat until the patient stops to survive. It should be noted that the lengths of each of these intervals may differ.

The number of repetitions is an objective of the study; the lower the number, the better the treatment administered. Is it enough to assess the credibility of treatments? Perhaps No. We can also consider the number of recurrences along with total duration of wellness of a patient; the longer the duration of wellness, the better the treatment. In our study, we have compared the above two methods: one that only considers the number of recurrences of the disease versus the one that considers both the number of recurrences and the duration of wellness.

We have taken into account a patient's immunity when estimating survival probabilities and hazard rates. We have also conducted statistical tests for the same in both parametric(6) and non-parametric(7.3) setup.

2 Duration of Wellness

Let X_j = Duration of wellness after j^{th} occurrence of a chronic disease, $j = 1, 2, 3, \dots$. Assume $X_j \sim \text{Gamma}(\alpha, p) \quad \forall j = 1, 2, \dots$ independently.

Then $Y_N = \sum_{j=1}^N X_j$ = total time of wellness after onset of the chronic disease till N^{th} recurrence. Here, N is assumed to be random following some count data distribution, say, $P(\lambda)$ distribution.

2.1 Distribution of Y_N

$$\begin{aligned}
P[Y_N \leq y] &= P \left[\sum_{j=1}^N X_j \leq y \right] = E_N P \left[\sum_{j=1}^N X_j \leq y \mid N \right] \\
&= E_N P[Y_N \leq y \mid N] \quad \text{where } Y_N \mid N \sim \text{Gamma}(\alpha, Np) \\
&= E_N \int_0^y \frac{\alpha^{Np}}{\Gamma(Np)} e^{-\alpha x} x^{Np-1} dx \\
&\approx \int_0^y \alpha e^{-\alpha x} E_N \left\{ \frac{(\alpha x)^{Np-1}}{(Np-1)!} \right\} dx, \quad \text{assuming } Np \text{ integer.} \\
\Rightarrow f_{Y_N}(y) &\approx \alpha e^{-\alpha y} E_N \left\{ \frac{(\alpha y)^{Np-1}}{(Np-1)!} \right\}, \quad y > 0. \\
&= \alpha e^{-\alpha y} \Psi(\alpha, p, \lambda, y).
\end{aligned}$$

(i) **Survival function of Y_N :**

$$S_{Y_N}(t) = P[Y_N > t] = \int_t^\infty f_{Y_N}(y) dy \approx \int_t^\infty \alpha e^{-\alpha y} E_N \left\{ \frac{(\alpha y)^{Np-1}}{(Np-1)!} \right\} dy$$

(ii) **Conditional time to event probability at x given $Y_N > t$:**

$$\begin{aligned}
C_{Y_N}(x, t) &= P[Y_N > x + t \mid Y_N > t] \\
&= \frac{\int_{x+t}^\infty \alpha e^{-\alpha u} E_N \left\{ \frac{(\alpha u)^{Np-1}}{(Np-1)!} \right\} du}{\int_t^\infty \alpha e^{-\alpha u} E_N \left\{ \frac{(\alpha u)^{Np-1}}{(Np-1)!} \right\} du} \\
&= \frac{\int_{x+t}^\infty \alpha e^{-\alpha u} E_N \left\{ \frac{(\alpha u)^{Np-1}}{(Np-1)!} \right\} du}{S_{Y_N}(t)}
\end{aligned}$$

(iii) Hazard rate / Failure rate of Y_N :

$$h_{Y_N}(t) = \frac{f_{Y_N}(t)}{S_{Y_N}(t)} = \frac{f_{Y_N}(t)}{\int_t^\infty f_{Y_N}(y) dy} = \frac{\alpha e^{-\alpha t} E_N \left\{ \frac{(\alpha t)^{Np-1}}{(Np-1)!} \right\}}{\int_t^\infty \alpha e^{-\alpha y} E_N \left\{ \frac{(\alpha y)^{Np-1}}{(Np-1)!} \right\} dy}$$

2.2 Distribution of T

Let C be the random censoring variable and δ be the censoring indicator

$$\text{Let } T = \begin{cases} Y_N & \text{if } Y_N < C \\ C & \text{if } Y_N \geq C \end{cases}$$

$$\therefore \delta = \begin{cases} 1 & \text{if } Y_N < C \quad (\text{uncensored case}) \\ 0 & \text{if } Y_N \geq C \quad (\text{censored case}) \end{cases}$$

$$\Rightarrow T = \min\{Y_N, C\}$$

We have assumed, Y_N to be independent of C .

(i) Survival function of T :

$$S_T(t) = P(T > t) = P(Y_N > t) \cdot P(C > t) = S_{Y_N}(t) \bar{F}_C(t).$$

(ii) Conditional time to event probability at x given $T > t$:

$$C_T(x, t) = P(T > x + t \mid T > t) = \frac{P(Y_N > x + t, C > x + t)}{P(T > t)} = \frac{S_{Y_N}(x + t) \bar{F}_C(x + t)}{S_T(t)}$$

(iii) Hazard rate / Failure rate of T

To find h_T , we need to find f_T :

$$\begin{aligned}
f_T(t) &= \frac{d}{dt} F_T(t) \\
&= -\frac{d}{dt} \{P(Y_N > t) \cdot P(C > t)\} \\
&= -\frac{d}{dt} \{S_{Y_N}(t) \cdot \bar{F}_C(t)\} \\
&= -\left\{ S_{Y_N}(t) \frac{d}{dt} \bar{F}_C(t) + \bar{F}_C(t) \cdot \frac{d}{dt} S_{Y_N}(t) \right\} \\
&= S_{Y_N}(t) \left\{ -\frac{d}{dt} \bar{F}_C(t) \right\} + \bar{F}_C(t) \cdot \left\{ -\frac{d}{dt} S_{Y_N}(t) \right\} \\
&= S_{Y_N}(t) \cdot f_C(t) + \bar{F}_C(t) \cdot f_{Y_N}(t). \\
\therefore h_T(t) &= \frac{f_T(t)}{S_T(t)} = \frac{f_C(t)}{\bar{F}_C(t)} + \frac{f_{Y_N}(t)}{S_{Y_N}(t)} = h_C(t) + h_{Y_N}(t)
\end{aligned}$$

2.3 Likelihood function

The likelihood function of α and λ (for given p) denoted by $L()$ is:

$$L(\alpha, \lambda \mid \mathbf{t}) = \prod_{i=1}^n \{f_{Y_N}(t_i) \cdot \bar{F}_C(t_i)\}^{\delta_i} \{S_{Y_N}(t_i) \cdot f_C(t_i)\}^{1-\delta_i}$$

Therefore, the log-likelihood denoted by $\ell()$ is:

$$\ell(\alpha, \lambda \mid \mathbf{t}) = \sum_{i=1}^n [\delta_i \{\ln f_{Y_N}(t_i) + \ln \bar{F}_C(t_i)\} + (1 - \delta_i) \{\ln S_{Y_N}(t_i) + \ln f_C(t_i)\}]$$

Suppose $C \sim \text{Exp} \left(\text{mean} = \frac{1}{\phi} \right)$, independently of Y_N .

$$\Rightarrow f_C(t) = \phi e^{-\phi t}, \quad \bar{F}_C(t) = e^{-\phi t}$$

Then,

$$\begin{aligned}
\ell(\alpha, \lambda \mid \mathbf{t}) &= \sum_{i=1}^n \delta_i \left\{ \ln \alpha e^{-\alpha t_i} \Psi + \ln e^{-\phi t_i} \right\} + (1 - \delta_i) \left\{ \ln \int_{t_i}^{\infty} \alpha e^{-\alpha x} \Psi \, dx + \ln \phi e^{-\phi t_i} \right\} \\
&= \sum_{i=1}^n \delta_i \{ \ln \alpha - \alpha t_i + \ln \Psi - \phi t_i \} + (1 - \delta_i) \left\{ \ln \int_{t_i}^{\infty} \alpha e^{-\alpha x} \Psi \, dx + \ln \phi - \phi t_i \right\}
\end{aligned} \tag{1}$$

In particular for $p = 1$, $X_j \sim \text{Gamma}(\alpha, 1) \quad \forall j = 1, 2, \dots$

$$\begin{aligned}
\frac{\partial \ell}{\partial \phi} &= \sum_{i=1}^n \left[\delta_i \{-t_i\} + (1 - \delta_i) \left\{ \frac{1}{\phi} - t_i \right\} \right] \\
&= \sum_{i=1}^n \frac{(1 - \delta_i)}{\phi} - \sum_{i=1}^n t_i
\end{aligned}$$

Setting $\frac{\partial \ell}{\partial \phi} = 0$ we get MLE of ϕ :

$$\hat{\phi} = \frac{\sum_{i=1}^n (1 - \delta_i)}{\sum_{i=1}^n t_i} = \frac{\text{Total no. of censored observations}}{\sum_{i=1}^n t_i}. \tag{2}$$

2.4 Simulation

We take $n = 100$ patients for our study.

For different choices of $\lambda = 1.5, 1.7, 1.8, 2.0, 2.2, 2.5$, we simulate N_i from $P(\lambda)$, $i = 1, 2, \dots, n$. Similarly, for the different choices of $\alpha = 0.2, 0.5, 0.9, 1.3, 1.5$ we simulate Y_N

from the pdf of Y_N using Acceptance-Rejection Algorithm. ¹

We have chosen $g \sim \text{Exp}(\alpha)$ and $M = 2$.

Thus we get T_i for every $i = 1, \dots, n$.

Method of estimation of parameters

To find MLE of α and λ , we had maximized (1) with respect to (α, λ) . For this we take an interval for both alpha and lambda containing true values of them. Now we locally maximize the likelihood for each pair of (α, λ) simultaneously. Now, we repeat the above algorithm for $m = 100$ times to obtain the MSE of the estimators as illustrated in following table:

Table 1: True Values, Estimated Values, and Mean Squared Errors (MSE)

α, λ, ϕ	$\hat{\alpha}, \hat{\lambda}, \hat{\phi}$	$\text{MSE}(\hat{\alpha}, \hat{\lambda}, \hat{\phi})$
0.2, 1.5, 0.10	0.1, 0.5, 0.12	0.002, 0.013, 0.002
0.5, 1.7, 0.15	0.3, 0.5, 0.15	0.013, 0.015, 0.008
0.5, 2.0, 0.20	0.4, 1.5, 0.18	0.014, 0.015, 0.004
0.9, 2.2, 0.15	0.9, 1.5, 0.17	0.007, 0.007, 0.001
1.3, 1.8, 0.13	0.9, 1.5, 0.14	0.018, 0.026, 0.001
1.5, 2.5, 0.10	1.2, 1.5, 0.14	0.020, 0.024, 0.001

From the above table, we can see that the MSE of parameters are small but the estimates of λ are far from the true value of λ . Here $\hat{\lambda}$ always takes the lower bound of the interval of λ . We have illustrated it through the following graphs(a and l denote the α and λ respectively).

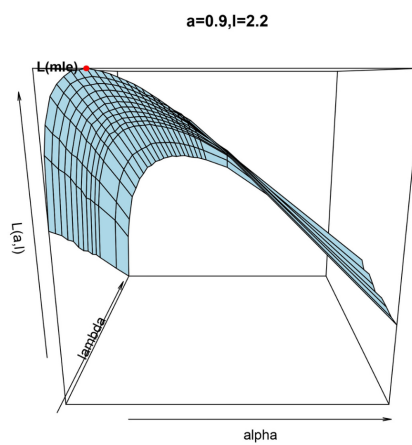
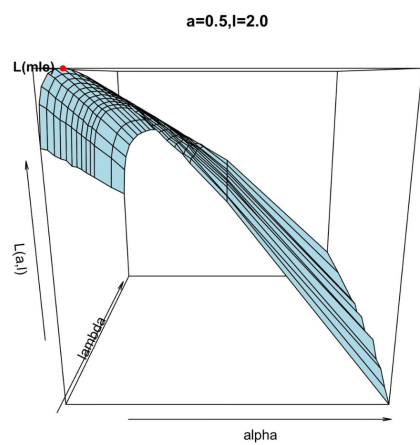
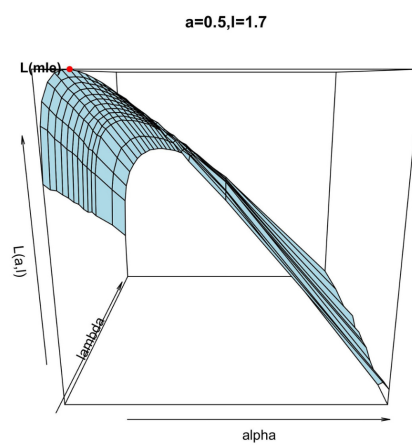
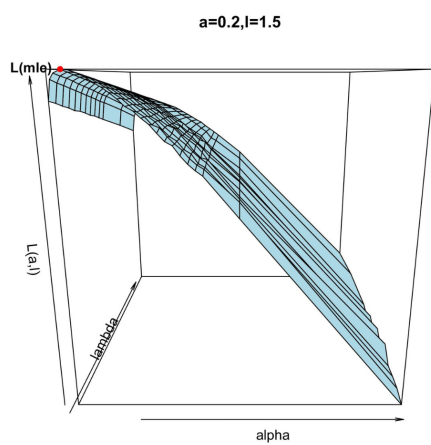
¹To generate a single sample X from $f(x)$:

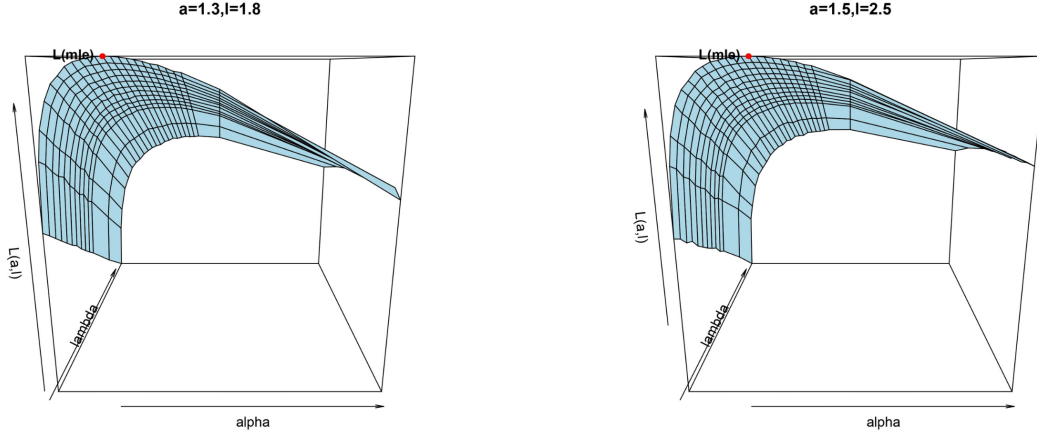
1. Generate a random sample X from the proposal distribution $g(x)$.
2. Generate a uniform random variable U from the standard uniform distribution: $U \sim \text{Uniform}(0, 1)$.
3. Check the following condition:

$$U \leq \frac{f(X)}{M \cdot g(X)}$$

4. Accept or Reject:

- If True: Accept X as a valid sample from $f(x)$.
- If False: Reject X and return to Step 1 (repeat until a candidate is accepted).





The above graphs show that the likelihood function is a smooth curved surface which peaks for the lower bound of the interval of λ irrespective of the true value of λ . This method of estimation did not provide an estimate of λ close to its true value. Hence, we substitute λ and ϕ by average number of recurrences of the disease and $\hat{\phi}$ from (2) respectively to get the likelihood in terms of α which is maximized w.r.t α to obtain the MLE of α . To maximize the likelihood w.r.t α we take an interval containing true value of α and calculate the likelihood value for each of the points in that interval and take that α for which the likelihood gets maximized locally in that interval.

2.5 Checking normality of parameters

For a given true value of (α, λ, ϕ) , simulate its mle $(\hat{\alpha}, \hat{\lambda}, \hat{\phi})$ for m times. Let $(\hat{\alpha}_i, \hat{\lambda}_i, \hat{\phi}_i)$ denote the mle at i^{th} repetition, $i = 1, 2, \dots, m$

$$\text{If } \sqrt{n} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\lambda} - \lambda \\ \hat{\phi} - \phi \end{pmatrix} \xrightarrow{\mathcal{D}} N_3(\mathbf{0}, i^{-1}(\boldsymbol{\theta})), \quad i(\boldsymbol{\theta})^{3 \times 3} = \text{Fisher information matrix.}$$

where $\boldsymbol{\theta} = (\alpha, \lambda, \phi)'$.

The exponent is given by Q

$$Q = n \left(\hat{\alpha} - \alpha, \hat{\lambda} - \lambda, \hat{\phi} - \phi \right) i(\boldsymbol{\theta}) \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\lambda} - \lambda \\ \hat{\phi} - \phi \end{pmatrix} \xrightarrow{\mathcal{D}} \chi_3^2 \text{ (approx.)}$$

$$Q_i = n \left(\hat{\alpha}_i - \alpha, \hat{\lambda}_i - \lambda, \hat{\phi}_i - \phi \right) i(\boldsymbol{\theta}) \begin{pmatrix} \hat{\alpha}_i - \alpha \\ \hat{\lambda}_i - \lambda \\ \hat{\phi}_i - \phi \end{pmatrix} \quad \forall i = 1(1)m.$$

where

$$i^{-1}(\boldsymbol{\theta}) = \begin{pmatrix} V(\hat{\alpha}) & \text{Cov}(\hat{\alpha}, \hat{\lambda}) & \text{Cov}(\hat{\alpha}, \hat{\phi}) \\ & V(\hat{\lambda}) & \text{Cov}(\hat{\lambda}, \hat{\phi}) \\ & & V(\hat{\phi}) \end{pmatrix} = \Sigma, \text{ say}$$

$$\hat{\Sigma} = \begin{pmatrix} \hat{V}(\hat{\alpha}) & \hat{\text{Cov}}(\hat{\alpha}, \hat{\lambda}) & \hat{\text{Cov}}(\hat{\alpha}, \hat{\phi}) \\ & \hat{V}(\hat{\lambda}) & \hat{\text{Cov}}(\hat{\lambda}, \hat{\phi}) \\ & & \hat{V}(\hat{\phi}) \end{pmatrix}$$

3 Number of Recurrences

Let N denote the number of recurrences of a chronic disease. Clearly, N is a non-negative random variable. Assume $N \sim \text{Poisson}(\lambda)$, $\lambda \in (0, \infty)$. Here, we do not take into account the duration of wellness.

3.1 Distribution of N

$$\begin{aligned} P(N \leq t) &= \sum_{i=0}^{[t]} P(N = i) \\ &= \sum_{i=0}^{[t]} \frac{e^{-\lambda} \lambda^i}{i!} \end{aligned}$$

(i) Survival function of N

$$\begin{aligned} S_N(t) &= P(N > t) = 1 - P(N \leq t) \\ &= 1 - \sum_{i=0}^{[t]} \frac{e^{-\lambda} \lambda^i}{i!} \end{aligned}$$

(ii) Conditional time to event probability at x given $N > t$

$$\begin{aligned} C_N(x, t) &= P(N > x + t \mid N > t) = \frac{P(N > x + t, N > t)}{P(N > t)} \\ &= \frac{P(N > x + t)}{P(N > t)} \\ &= \frac{S_N(x + t)}{S_N(t)} = \frac{1 - \sum_{i=0}^{[x+t]} \frac{e^{-\lambda} \lambda^i}{i!}}{1 - \sum_{i=0}^{[t]} \frac{e^{-\lambda} \lambda^i}{i!}} \end{aligned}$$

(iii) Hazard rate / Failure rate of N :

$$\begin{aligned} h_N(t) &= P(N = [t] + 1 \mid N \geq [t] + 1) = \frac{P(N = [t] + 1)}{P(N \geq [t] + 1)}, \text{ for } t \geq 1 \\ &= \frac{\frac{e^{-\lambda} \lambda^{[t]+1}}{([t] + 1)!}}{\sum_{i=[t]+1}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!}} \end{aligned}$$

3.2 Distribution of T^*

Let C^* be the random censoring variable and ν be the censoring indicator.

$$\text{Let } T^* = \begin{cases} N & \text{if } N < C^* \\ C^* & \text{if } N \geq C^* \end{cases} \quad \nu = \begin{cases} 1 & \text{if } N < C^* \quad (\text{uncensored case}) \\ 0 & \text{if } N \geq C^* \quad (\text{censored case}) \end{cases}$$

$$\Rightarrow T^* = \min\{N, C^*\}$$

We have assumed N to be independent of C^* .

Let $C^* \sim \text{Poisson}(k), k \in (0, \infty)$

(i) Survival Function of T^*

Using the independence of N and C^* , the survival function $S_{T^*}(t)$ is the product of their individual survival functions:

$$\begin{aligned} S_{T^*}(t) &= P(T^* > t) = P(\min\{N, C^*\} > t) \\ &= P(N > t \text{ and } C^* > t) \\ &= P(N > t) \cdot P(C^* > t) \\ &= \left(1 - \sum_{j=0}^{[t]+1} \frac{e^{-\lambda} \lambda^j}{j!}\right) \left(1 - \sum_{j=0}^{[t]+1} \frac{e^{-k} k^j}{j!}\right) \end{aligned}$$

(ii) **Conditional time to event probability at x given $T^* > t$**

$$\begin{aligned}
C_{T^*}(x, t) &= P(T^* > x + t \mid T^* > t) = \frac{P(T^* > x + t, T^* > t)}{P(T^* > t)} \\
&= \frac{P(T^* > x + t)}{P(T^* > t)} \\
&= \frac{S_{T^*}(x + t)}{S_{T^*}(t)} \\
&= \frac{\left(1 - \sum_{j=0}^{[x+t]+1} \frac{e^{-\lambda} \lambda^j}{j!}\right) \left(1 - \sum_{j=0}^{[x+t]+1} \frac{e^{-k} k^j}{j!}\right)}{\left(1 - \sum_{j=0}^{[t]+1} \frac{e^{-\lambda} \lambda^j}{j!}\right) \left(1 - \sum_{j=0}^{[t]+1} \frac{e^{-k} k^j}{j!}\right)}
\end{aligned}$$

(iii) **Hazard Function of T^***

$$\begin{aligned}
h_N(t) &= \frac{P(N = [t] + 1)}{P(N \geq [t] + 1)} \quad \forall t \geq 1 \\
h_{T^*}(t) &= \frac{P[T^* = [t] + 1]}{P[T^* \geq [t] + 1]} \\
P[T^* = [t] + 1] &= P[\min\{N, C^*\} = \underbrace{[t] + 1}_K] \\
&= P[N = K, C^* > K] + P[C^* = K, N \geq K] \\
&= P[N = K]P[C^* > K] + P[C^* = K]P[N \geq K] \\
\therefore h_{T^*}(t) &= \frac{P[N = K]P[C^* > K]}{P[N \geq K]P[C^* \geq K]} + \frac{P[C^* = K]}{P[C^* \geq K]} \\
h_{T^*}(t) &= h_N(t) \cdot \frac{P[C^* > K]}{P[C^* \geq K]} + h_{C^*}(t) \quad \forall t \geq 1
\end{aligned}$$

3.3 Likelihood function

The log-likelihood is given by

$$\begin{aligned}\ell(k, \lambda \mid \mathbf{t}) &= \sum_{i=1}^m \nu_i \{ \ln f_N(t_i) + \ln \bar{F}_{C^*}(t_i) \} + (1 - \nu_i) \{ \ln S_N(t_i) + \ln f_{C^*}(t_i) \} \\ &= \sum_{i=1}^m \nu_i \left\{ \ln \frac{e^{-\lambda} \lambda^{t_i}}{t_i!} + \ln \left(1 - \sum_{j=0}^{t_i} \frac{e^{-k} k^j}{j!} \right) \right\} + (1 - \nu_i) \left\{ \ln S_N(t_i) + \ln \frac{e^{-k} k^{t_i}}{t_i!} \right\}\end{aligned}\tag{3}$$

3.4 Simulation

We take $n = 100$ patients for our study.

For a true value of λ we simulate N_i from $P(\lambda)$, $i = 1, 2, \dots, n$ to obtain $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n n_i$.

We obtain the MLE of k similarly as we obtain in case of α by substituting λ by $\hat{\lambda}$ in (3).

4 Immunity as a covariate

Now this is true that for any patient, how many times the disease will recur, should depend on the immunity of the patient. Let u denote the immunity of a patient and for simplicity, suppose u is ordinal categorical variable having 3 ordinal categories of immunity $u = 0$ (low), 1 (moderate), 2 (high). Also this seems to be quite rational that higher u tends to result in lower recurrence no. of the disease on an average, i.e.,

$$N|u \sim P(\lambda(u)), \text{ where } \lambda(u) \downarrow u$$

$$\text{Take } \lambda(u) = \lambda a^u, \lambda > 0, a \in (0, 1)$$

Hence, accordingly one can update the distribution of Y_N as follows:

$$Y_N = \sum_{j=1}^N X_j \text{ where } X_j \sim \text{Gamma}(\alpha(u), p) \forall j = 1, 2, \dots \text{ independently}$$

Take $\alpha(u) = \frac{\alpha}{1+u} \downarrow u$ and $N|u \sim P(\lambda(u))$.

Since with increase in immunity, on an average the total duration of wellness upto N^{th} recurrence should increase. Hence, we have taken the above choice of $\alpha(u)$.

Also the distribution of X_j is independent of N .

$$L(\lambda|\mathbf{u}) = \prod_{i=1}^n \frac{e^{-\lambda a^{u_i}} (\lambda a^{u_i})^{n_i}}{n_i!} = \frac{e^{-\lambda \sum_{i=1}^n a^{u_i}} \lambda^{\sum_{i=1}^n n_i} a^{\sum_{i=1}^n n_i u_i}}{(\prod_{i=1}^n n_i!)} \quad \text{where } N_i|u_i \sim P(\lambda(u_i) = \lambda a^{u_i})$$

$\Rightarrow$ log likelihood:

$$\ell(\lambda|\mathbf{u}) = -\lambda \sum_{i=1}^n a^{u_i} + \left(\sum_{i=1}^n n_i \right) \ln \lambda + \text{term independent of } \lambda$$

$$\frac{\partial \ell}{\partial \lambda} = -\sum_{i=1}^n a^{u_i} + \frac{\sum_{i=1}^n n_i}{\lambda} = 0 \implies \hat{\lambda} = \frac{\sum_{i=1}^n n_i}{\sum_{i=1}^n a^{u_i}} \quad (4)$$

We get the respective survival, conditional survival and hazard for different levels of u by the above definitions.

Simulation

For each patient we simulate their corresponding immunity value which is done by taking a random sample with replacement of size n from 0, 1, 2.

For choosing $a = 0.9$, we calculate $\lambda(u) = \lambda a^u$ for every patient.

Then we follow the exact steps for simulation in (2.4) and (3.4) by replacing λ by $\hat{\lambda}(u)$ from (4) and α by $\alpha(u)$ in (1) and (3).

We repeat the algorithm for $m = 100$ times to obtain the MSE of the estimators as illustrated in following table.

Table 2: True Values, Estimated Values and Mean Squared Errors (MSE)

Parameter	True Value	Estimated Value ($\hat{\cdot}$)	MSE of Estimator
α	0.50	0.7141	0.05184542
λ	1.70	2.1668	0.24544670
ϕ	0.15	0.1506	0.00040709
k	3.00	3.0050	0.05661936

Observation on normality of parameters

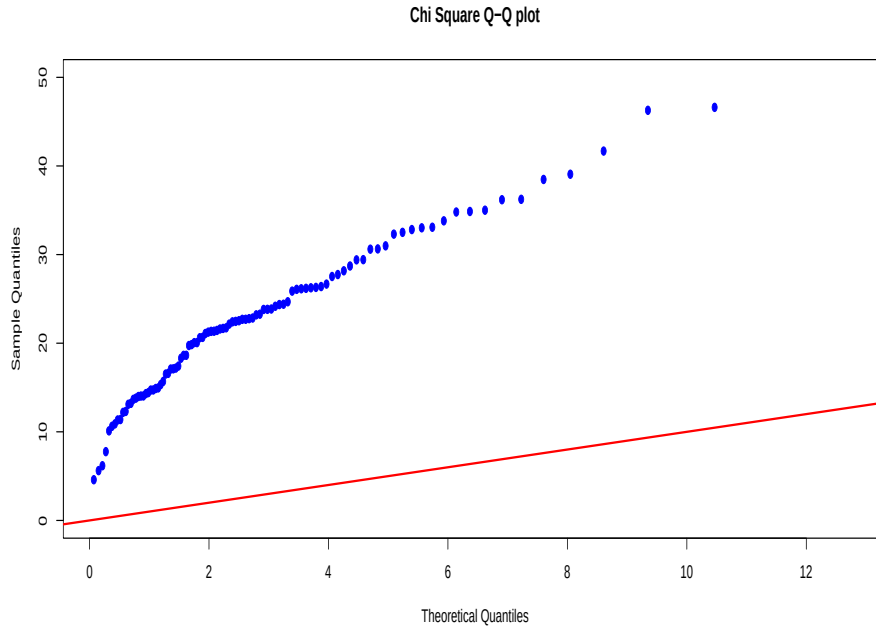


Figure 3: Chi-square Q-Q Plot for exponent

Remark: The datapoints are far from the $y = x$ line indicating that the parameters do not follow trivariate normal distribution.

5 Comparison of T and T^*

5.1 Survival functions

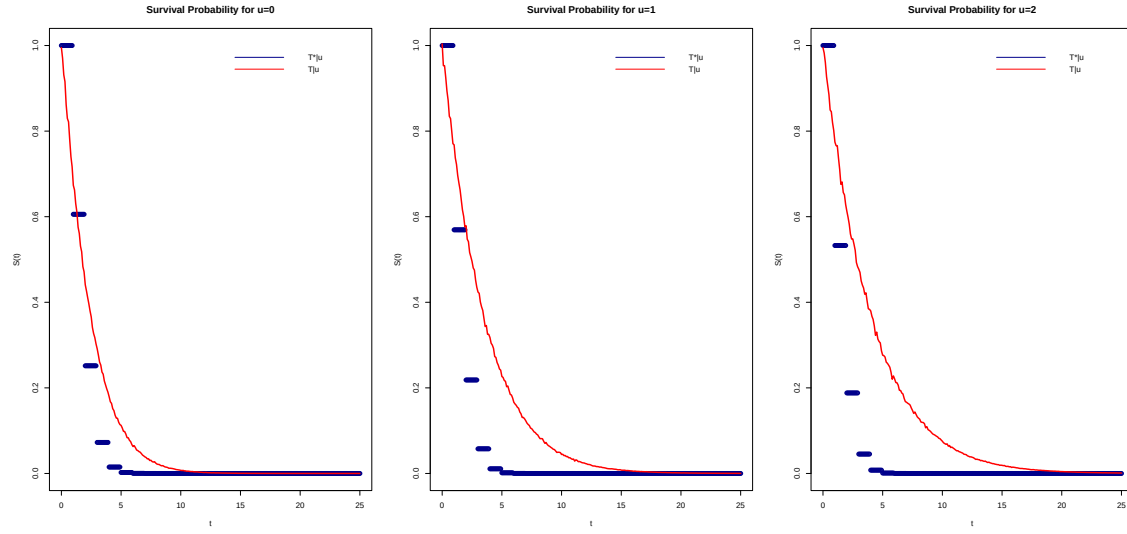


Figure 4: Survival function for different levels of immunity

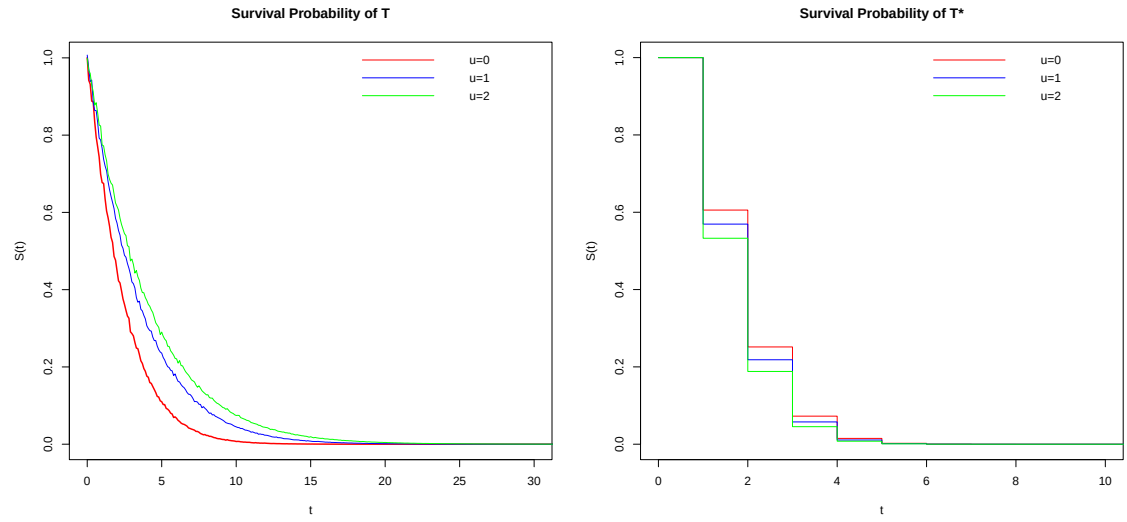


Figure 5: Survival function for T and T^*

Observations:

1. From the graph of T , we can see that as immunity increases, the survival probability increases. Also, the curve decreases more sharply for lower values of immunity early on. This indicates that lower the immunity, greater is the risk of recurrence of the disease.
2. From the graph of T^* , we cannot differentiate the effect of immunity on the survival curves. This is due to the shortcomings of the model that includes only N , i.e., the number of recurrences in comparison to the one which includes both Y_N and N .
3. Moreover, from an overall point of view, the curve of S_T always lies above that of S_{T^*} for every level of immunity, i.e., for a given t , the probability of survival of a patient when considering both Y_N and N is higher than the probability of survival of a patient when considering only N .

5.2 Conditional survival functions

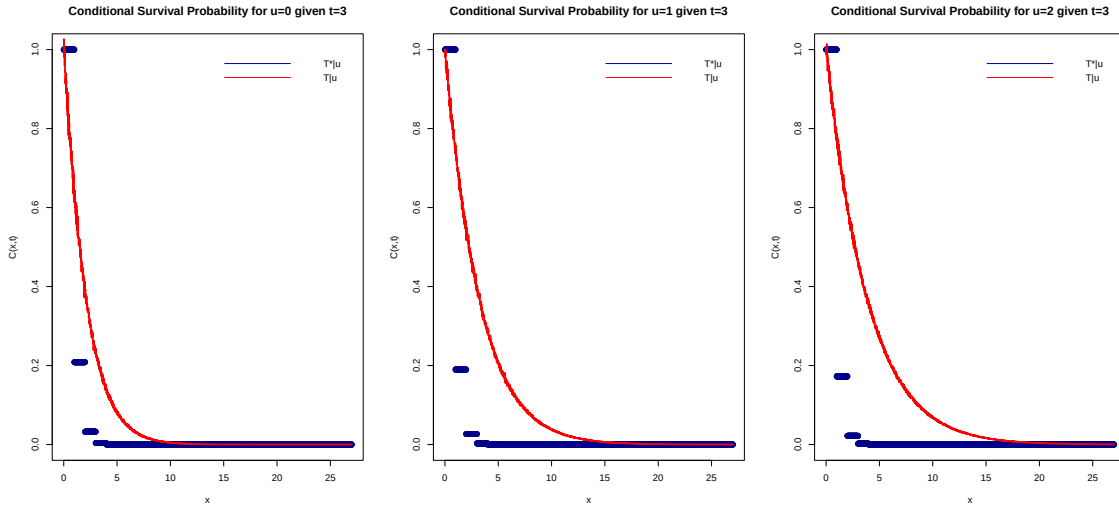


Figure 6: Conditional survival function for different levels of immunity

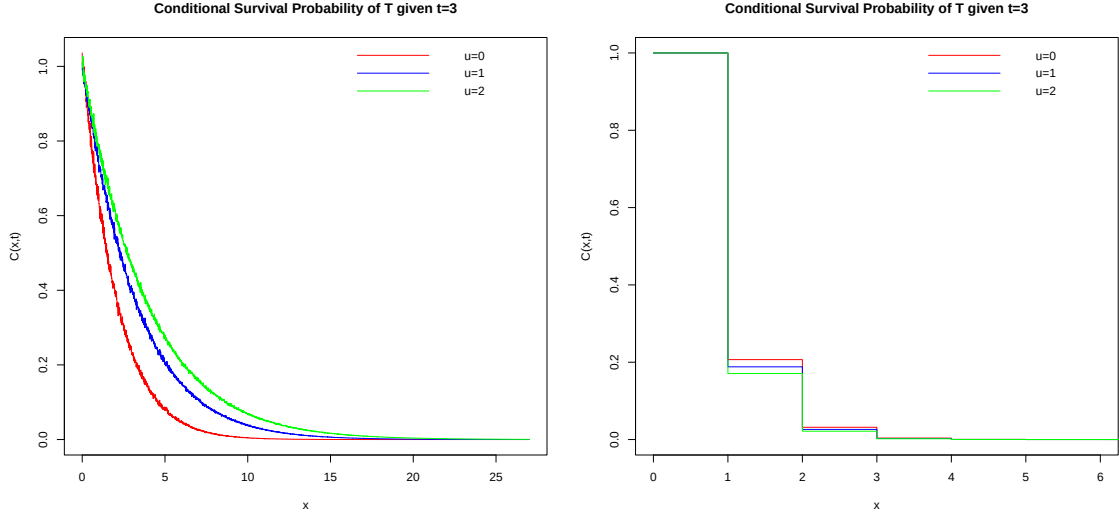


Figure 7: Conditional survival function for T and T^*

Observations:

1. From the graph of T , we can see that given the total duration of wellness for a patient exceeds $t=3$ units, the probability that they continue to remain in a state of well-being for additional x units of time increases with increase in immunity.
2. From the graph of T^* , for a patient, given that the number of recurrence of the disease exceeds 3 times, the probability that the disease will occur again decreases with increase in immunity.
3. Moreover, the curve of C_T usually lies above that of C_{T^*} for every level of immunity.

5.3 Hazard rates

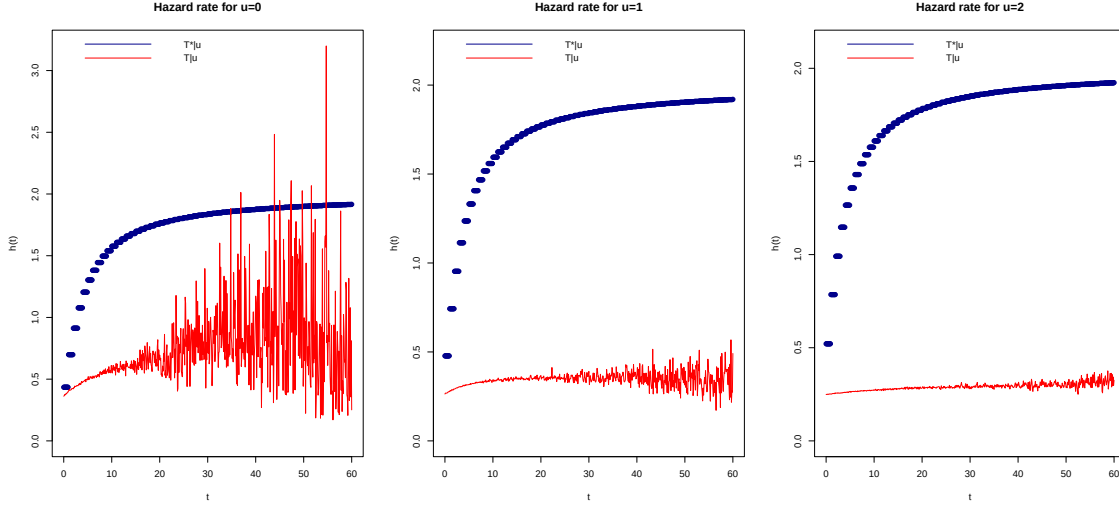


Figure 8: Hazard rates for different levels of immunity

Observations:

1. From the graph of T , we can see that given that a patient remains well, the chance of the disease recurring immediately after decreases with increase in immunity. For low immunity patients, the hazard rate fluctuates heavily which is the fallacy of the model without immunity. With increase in immunity the band width of the fluctuation decreases.
2. The curve of T^* , increases rapidly for smaller values of t and eventually stabilizes.
3. Overall, the gap between h_T and h_{T^*} increases significantly with increase in immunity.

6 Test of equality of survival functions

Let $S_1(t)$ denote the survival function of T and $S_2(t)$ denote the survival function of T^*

To test $H_0 : S_1(t) = S_2(t)$ versus $H_1 : S_1(t) > S_2(t)$

$$V\left(\hat{S}_1(t)\right) \approx \frac{S_1(t)(1-S_1(t))}{n} = \sigma_1^2, \quad \text{say.}$$

$$V\left(\hat{S}_2(t)\right) \approx \frac{S_2(t)(1-S_2(t))}{n} = \sigma_2^2, \quad \text{say.}$$

Under $H_0 : \sigma_1^2 = \sigma_2^2$

$$\begin{aligned} \frac{S_1(t)(1-S_1(t))}{n} &= \frac{S_2(t)(1-S_2(t))}{n} \\ &= \frac{S(t)(1-S(t))}{n} \simeq \frac{\hat{S}(t)(1-\hat{S}(t))}{n} \end{aligned}$$

$$\text{where } \hat{S}(t) = \frac{\hat{S}_1(t) + \hat{S}_2(t)}{2} \quad [\text{under } H_0, S_1(t) = S_2(t) = S(t)]$$

Let $\psi(S(t)) = \ln \frac{S(t)}{1-S(t)}$

By Delta method,

$$\begin{aligned} V\left(\psi(\hat{S}(t))\right) &= \left(\frac{\partial}{\partial \hat{S}(t)} \psi(\hat{S}(t))\right)_{\hat{S}(t)=S(t)}^2 V\left(\hat{S}(t)\right) \\ &= \left(\frac{1-\hat{S}(t)}{\hat{S}(t)} \times \frac{(1-\hat{S}(t)) - \hat{S}(t)(-1)}{(1-\hat{S}(t))^2}\right)_{\hat{S}(t)=S(t)}^2 \times \frac{S(t)(1-S(t))}{n} \\ &= \frac{1}{(S(t))^2(1-S(t))^2} \times \frac{S(t)(1-S(t))}{n} \\ &= \frac{1}{n S(t)(1-S(t))} \end{aligned}$$

So we get,

$$\ln \frac{\hat{S}_i(t)}{1 - \hat{S}_i(t)} \xrightarrow{\mathcal{D}} N \left(\ln \frac{S_i(t)}{1 - S_i(t)}, \frac{1}{n S_i(t)(1 - S_i(t))} \right), i = 1, 2$$

Under $H_0 : S_1(t) = S_2(t) = S(t)$, say

$$\ln \frac{\hat{S}_1(t)}{1 - \hat{S}_1(t)} - \ln \frac{\hat{S}_2(t)}{1 - \hat{S}_2(t)} \xrightarrow{\mathcal{D}} N \left(0, \frac{2}{n S(t)(1 - S(t))} \right)$$

$$Z_0 = \frac{\left\{ \ln \frac{\hat{S}_1(t)}{1 - \hat{S}_1(t)} - \ln \frac{\hat{S}_2(t)}{1 - \hat{S}_2(t)} \right\} / \sqrt{2}}{\sqrt{\frac{1}{n \hat{S}(t)(1 - \hat{S}(t))}}} \xrightarrow{\mathcal{D}} N(0, 1).$$

We reject H_0 at level α if $Z_0 > \tau_\alpha$, upper α^{th} point of $N(0, 1)$.

For $\alpha = 0.05$,

Table 3: Proportion of rejection of the null hypothesis

Immunity	Acceptance Count	Rejection Count	Proportion(Reject)
Low	8	2	0.2
Moderate	1	9	0.9
High	0	10	1.0

Observation:

We can see that as immunity increases the proportion of rejection of the $H_0 : S_1(t) = S_2(t)$ by the above test increases which supports our deductions from exploratory analysis.

Remark:

The above test procedure is neither fully parametric nor non-parametric. We can say that this is a semi-parametric one since the value of Z_0 involves estimates of the parameters of the parent distribution but the construction of test and null distribution remains the same irrespective of the parent distribution of Y_N and N .

7 Non-parametric Setup

7.1 Estimation of $S(\cdot)$

Using Kaplan-Meier method of estimation of survival function, we get survival functions for different levels of u .

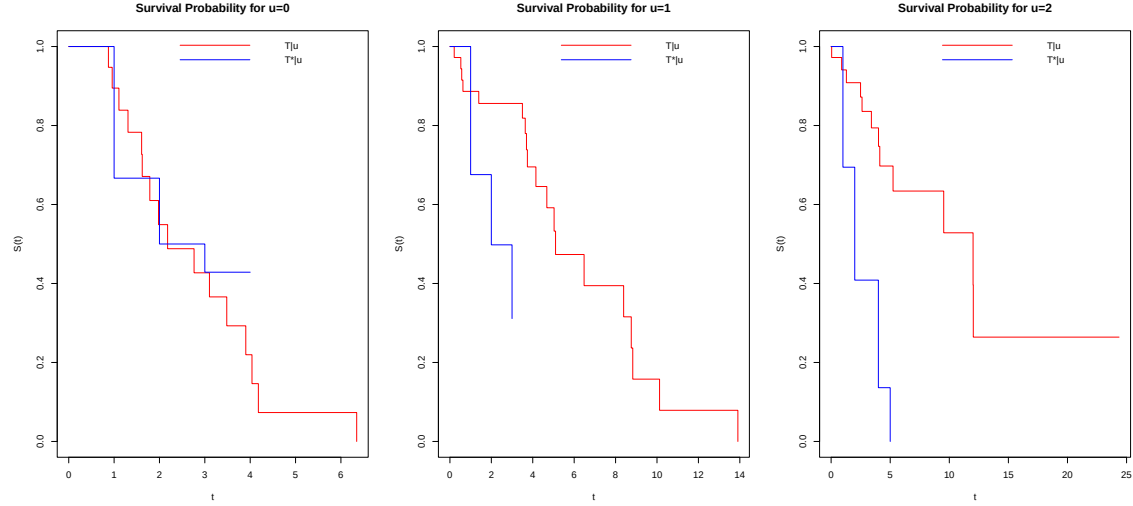


Figure 9: Survival functions for different levels of immunity

1. From the graph of T , we can see that as level of immunity increases the survival probability increases, i.e, the probability that a patient will remain in a state of well-being for a longer period of time increases. For $u=2$, the last observation is censored. Hence, the survival probability does not become 0.
2. From the graph of T^* , no noteworthy change is visible over covariate information. This shows that considering only number of recurrences is not enough to reflect the change in covariate values in the survival functions which indicates towards the inferiority of this model.
3. Moreover, the curve of S_T usually lies above that of S_{T^*} for every level of immunity. The gap between both the curves increase significantly with increase in level of immunity but for low immunity, i.e., no covariate effect case, there is no apparent difference between the two curves.

7.2 Estimation of $h(\cdot)$

Using Cox proportional hazard model and partial likelihood method of estimation, we get hazard functions for different levels of u . $h(t, u) = h_0(t)e^{\beta u}$, where $h_0(t)$ is the baseline hazard function independent of u . For low immunity, i.e. $u=0$, we get the baseline hazard function.

Estimate of β for T

This table shows the Cox proportional hazards regression results for T ($n = 100$, number of events = 52).

Table 4: Cox Proportional Hazards Model Results

Variable	coef	se(coef)	z	Pr(> z)	95% CI for exp(coef)
immunity	-0.7779	0.2047	-3.8	0.000144	[0.3076, 0.6861]

Likelihood ratio test = 15.59 on 1 df, $p = 8e-05$

Wald test = 14.44 on 1 df, $p = 1e-04$

Score (logrank) test = 15.51 on 1 df, $p = 8e-05$

Since p -value is too small, we can say that immunity is significant while modelling T .

Estimate of β for T^*

This table shows the Cox proportional hazards regression results for T^* ($n = 100$, number of events = 63).

Table 5: Cox Proportional Hazards Model Results

Variable	coef	se(coef)	z	Pr(> z)	95% CI for exp(coef)
immunity	0.03095	0.16727	0.185	0.853	[0.7431, 1.432]

Likelihood ratio test = 0.03 on 1 df, $p = 0.9$

Wald test = 0.03 on 1 df, $p = 0.9$

Score (logrank) test = 0.03 on 1 df, $p = 0.9$

Since p -value is large, we can say that immunity is insignificant while modelling T^* .

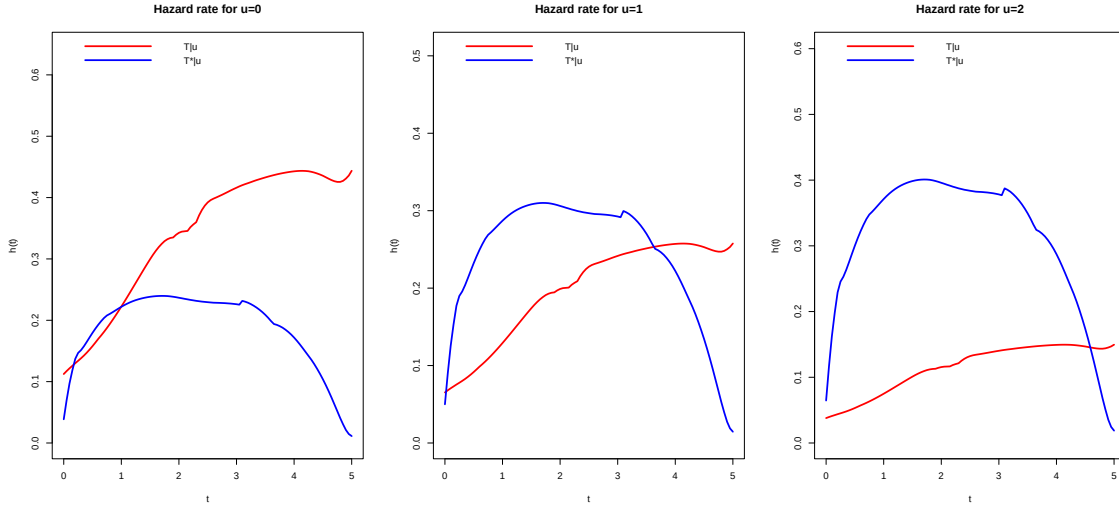


Figure 10: Hazard rates for different levels of immunity

1. From the graph of T , we can see that as level of immunity increases the hazard decreases.
2. From the graph of T^* , we can see that irrespective of immunity, the hazard curve initially increases and after a certain value of t it goes downwards resulting in an inverted U-shaped curve.
3.
 - For $u=0$, the curve of h_T lies above the curve of h_{T^*} . This is because while considering only N , we lack information as compared to the case involving Y_N . Given that there is no covariate information, apparently the risk of occurrence of the disease seems to be less which is misleading.
 - For $u=1$, the curve of h_T lies below the curve of h_{T^*} except for a few large values of t .
 - For $u=2$, the curve of h_T lies well below the curve of h_{T^*} and there is a significant gap between both the curves.

Remark: From exploratory analysis we can conclude that sometimes considering only N may be misleading in the sense that it does not show the severity of a disease due to lack of information.

7.3 Testing of Hypothesis(MH Test)

To test $H_0 : S_1(t) = S_2(t)$ vs. $H_1 : S_1(t) > S_2(t)$

Combine the two groups and arrange the $(n_1 + n_2)$ observation in ascending order.

Let τ_j be the j^{th} time point with some uncensored observation. (Ignore the time points where there are only censored observation)

Let n_{kj} = no. of individuals in k^{th} group ($k = 1, 2$) at τ_j ($n_{1j} + n_{2j} = n_j$)

m_{kj} = no. of individuals with $k = E$ or NE at τ_j

a_j = no. of individuals in 1^{st} group having E

$$E(a_j) = \frac{m_{1j} n_{1j}}{n_j}$$

$$V(a_j) = \frac{m_{1j} n_{1j} m_{2j} n_{2j}}{n_j^2(n_j - 1)}$$

$$MH = \frac{\sum_j (a_j - E(a_j))}{\sqrt{\sum_j \text{Var}(a_j)}} \xrightarrow{\mathcal{D}} N(0, 1) \Rightarrow (MH)^2 \xrightarrow{\mathcal{D}} \chi_{(1)}^2$$

1. Low Immunity Group

Group	N	Observed	Expected	$(O - E)^2/E$	$(O - E)^2/V$
group=1	31	21	24.2	0.425	1.31
group=2	31	21	17.8	0.578	1.31

Statistic: Chisq = 1.31 on 1 degree of freedom, $p = 0.3$

Since $p - \text{value} = 0.3$ is large we accept $H_0 : S_1(t) = S_2(t)$.

2. Moderate Immunity Group

Group	N	Observed	Expected	$(O - E)^2/E$	$(O - E)^2/V$
group=1	39	21	28	1.74	6.81
group=2	39	22	15	3.23	6.81

Statistic: Chisq = 6.81 on 1 degree of freedom, $p = 0.009$

Since $p - \text{value} = 0.009$ is small we reject $H_0 : S_1(t) = S_2(t)$.

3. High Immunity Group

Group	N	Observed	Expected	$(O - E)^2/E$	$(O - E)^2/V$
group=1	30	10	19	4.26	16.8
group=2	30	20	11	7.37	16.8

Statistic: Chisq = 16.8 on 1 degree of freedom, $p = 4 \times 10^{-5}$
 Since $p - value = 4 \times 10^{-5}$ is small we reject $H_0 : S_1(t) = S_2(t)$.

Observation:

In low immunity case the survival functions of T^* and T cannot be differentiated well from the graph but in case of moderate and high immunity groups there is significant difference between the survival functions. The test accepts the null in the first case(no covariate information) and rejects the null in the latter cases.

Hence, the confirmatory test results support the exploratory data analysis.

8 Applications

This kind of study may also be applicable in:

1. The study of the sleep cycle of an individual. The **duration of sleep**(Y_N) and the **number of times one feels sleepy during a day**(N) are random. Given one has a longer duration of sleep, the number of recurrence of sleepiness should decrease.
2. The **duration of inclusion of a player in a cricket team**(Y_N) and **number of times they get selected to play for the team**(N) are random. One might be interested in finding how long the player "survives" in the team.
3. The **duration of addiction to a substance**(Y_N) and **number of recurrence of urge to indulge in the addiction**(N) can be also be studied likewise. Stress level of an individual may be a potential covariate.

9 Limitations

1. Duration of wellness and number of recurrences are assumed to be independent. Intuitively, as number of recurrence increases, the duration of wellness of a patient is likely to decrease. Therefore, these two variables are highly dependent which has not been considered in our study.
2. Duration of wellness and duration of illness recur alternately. In our study we have not considered the latter. One may be interested to conduct a similar study by considering both the durations along with the number of recurrences of the chronic disease.
3. For further study, there is a scope of improvement using Bayesian analysis.

10 References

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3. Christopher Cox, Haitao Chu, Michael F. Schneider, Alvaro Muñoz, Parametric survival analysis and taxonomy of hazard functions for the generalized gamma distribution, Statistics in Medicine: Volume 26, Issue 23, Wiley, 2007, <https://doi.org/10.1002/sim.2836>
4. Sugata Sen Roy, Moumita Chatterjee, Estimating the hazard functions of two alternately occurring recurrent events, Journal of Applied Statistics, 2015, <http://dx.doi.org/10.1080/02664763.2014.1001329>

Appendix

The R code used in this study for analysis purpose is given below: <https://github.com/poddarprapti/A-Comparative-Performance-Study-on-Recurrent-Time-to-Event-Data>